A SHAI Subcase for the First Limit Bourgain–Rosenthal–Schechtman Space

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Status

This packet records a partial result for Johnson–Phillips–Schechtman's Problem from [5]. The source asks whether every complemented subspace of L^p, and in particular the Bourgain–Rosenthal–Schechtman examples, has the SHAI property. We prove the first limit BRS/Rosenthal subcase.

Theorem 1 (Main result)**.** *Let $1 < p < \infty$. The centered first limit Bourgain–Rosenthal–Schechtman space $R^{p,0}_\omega$ has the SHAI property. Consequently the non-centered first limit space*

$$R^p_\omega \cong \mathbb{K} \oplus R^{p,0}_\omega$$

also has the SHAI property.

This is a partial answer only: it does not settle arbitrary complemented subspaces of L^p, the unconditional-basis version, or all countable BRS spaces R^p_α.

Source Question

In [5, Problem 2.7], Johnson–Phillips–Schechtman ask:

> Suppose $p \in (1,\infty) \setminus \{2\}$ and let X be a complemented subspace of L^p. Does X have the SHAI property? What if, in addition, X has an unconditional basis? What if, in addition, X is one of the $\aleph_1$ spaces constructed in [1]?

Proof Intuition

The Johnson–Phillips–Schechtman proof for L^p, $1 < p < \infty$, uses a finite-dimensional decomposition with continuum many almost-disjoint coordinate subspaces, each isomorphic to the whole space. In the first limit BRS space the natural blocks are

$$H_n = L^0_p(\{0,1\}^n),$$

and a copy of H_m generally sits inside a later block H_k, $k \geq m$, rather than as a whole block. This is still enough. For every infinite set $A = \{a_1 < a_2 < \cdots\}$, embed H_m into H_{a_m} by ignoring the last $a_m - m$ Bernoulli coordinates. The resulting range is a complemented isometric copy of the whole first-limit BRS space.

Taking a continuum almost-disjoint family of infinite sets A gives continuum many self-copy projections P_A. If $A \cap B$ is finite, then $P_A P_B$ has finite-dimensional range, hence is finite rank. In any nonzero quotient of $\mathcal{L}(X)$, finite-rank operators vanish, so the images of the P_A's become pairwise orthogonal nonzero idempotents with uniformly bounded finite sums. A lower s-estimate inherited from L^p makes the associated decomposition in the quotient target boundedly complete, and the usual density contradiction follows.

Preliminaries

We use the concrete first-limit realization from [7]: let $\Omega_n = \{0,1\}^n$ with uniform probability and set

$$H_n = L_p^0(\Omega_n).$$

On the product probability space $\Omega = \prod_{n=1}^{\infty} \Omega_n$, let $J_n : H_n \to L^p(\Omega)$ be the n-th coordinate lifting. Then

$$X := R_\omega^{p,0} = \overline{\operatorname{span}}\{J_n(H_n) : n \in \mathbb{N}\}.$$

Each H_n is finite dimensional. The coordinate projection onto any subcollection of the exterior blocks is the restriction of a conditional expectation, hence has norm at most one.

We shall use the following standard estimate in the same way as [5, Section 1]. Since X is a subspace of an L^p-space, its canonical unconditional finite-dimensional decomposition has a disjoint lower s-estimate for some finite s, for instance $s = \max\{p, 2\}$: there is $C_s < \infty$ such that whenever $z_1, \dots, z_m \in X$ have disjoint exterior block supports,

$$\Big(\sum_{j=1}^{m} \|z_j\|^s \Big)^{1/s} \le C_s \Big\| \sum_{j=1}^{m} z_j \Big\|.$$

Block-Finite Self-Copies

Lemma 1 (Almost-disjoint complemented self-copies)**.** *There is a family $\mathcal{A} \subset [\mathbb{N}]^\omega$ of cardinality continuum and norm-one projections $P_A \in \mathcal{L}(X)$, $A \in \mathcal{A}$, such that:*

1. *$P_A X$ is isometric to X;*
2. *if $A \neq B$, then $P_A P_B$ is finite rank;*
3. *for every finite $\mathcal{F} \subset \mathcal{A}$ there are finite-rank operators R_A, $A \in \mathcal{F}$, such that*

$$\Phi(P_A) = \Phi(P_A - R_A)$$

for every homomorphism Φ whose kernel contains the finite-rank operators, and

$$\Big\| \sum_{A \in \mathcal{F}} (P_A - R_A) \Big\| \le 1.$$

Proof. Fix a continuum-sized almost-disjoint family $\mathcal{A} \subset [\mathbb{N}]^\omega$. For $A = \{a_1 < a_2 < \cdots\}$, we have $a_m \ge m$. Let $\rho_{m,A} : \Omega_{a_m} \to \Omega_m$ be the projection onto the first m Bernoulli coordinates. Pullback by $\rho_{m,A}$ gives an isometric embedding

$$u_{m,A} : H_m \longrightarrow H_{a_m}.$$

The conditional expectation from $L^p(\Omega_{a_m})$ onto the sigma algebra generated by those first m coordinates restricts to a norm-one projection from H_{a_m} onto $u_{m,A}H_m$.

Define $U_A : X \to X$ on finite sums by

$$U_A\Big(\sum_m J_m x_m\Big) = \sum_m J_{a_m} u_{m,A} x_m.$$

The summands remain independent and preserve their individual distributions, so U_A extends to an isometry. Define $V_A : X \to X$ by first projecting onto the exterior blocks a_m, then applying the above conditional expectation inside H_{a_m}, and finally identifying $u_{m,A}H_m$ with H_m. This is the restriction of a conditional expectation/factor map on the product probability space, so $\|V_A\| \le 1$, and $V_A U_A = I_X$. Put $P_A = U_A V_A$. Then P_A is a norm-one projection and $P_A X = U_A X$ is isometric to X.

If $A \neq B$, then $A \cap B$ is finite. The operator $P_A P_B$ can be nonzero only on exterior blocks indexed by $A \cap B$. Each such block is finite dimensional, hence $P_A P_B$ is finite rank.

Now let $\mathcal{F} \subset \mathcal{A}$ be finite and let

$$S = \bigcup_{\substack{A,B\in\mathcal{F}\\ A\neq B}} (A \cap B).$$

For $A \in \mathcal{F}$, let P_A^S be the same projection as P_A, but with all target exterior blocks in S deleted. Then $P_A - P_A^S$ has range contained in the finite-dimensional sum of the blocks H_k, $k \in S$, so it is finite rank. The remaining target block supports $A \setminus S$, $A \in \mathcal{F}$, are pairwise disjoint. Therefore $\sum_{A\in\mathcal{F}} P_A^S$ is again a coordinatewise conditional expectation onto mutually disjoint Bernoulli factors, and its norm is at most one. Taking $R_A = P_A - P_A^S$ proves the last assertion. □

Lemma 2 (Lower-estimate transfer for the self-copy projections). *Let $Q_1, \dots, Q_m$ be projections obtained from finitely many P_A's by deleting a common finite set of exterior target blocks as in the proof of Lemma 1, so that the ranges of the Q_j's have pairwise disjoint exterior block supports and $\|\sum_j Q_j\| \le 1$. If $T_1, \dots, T_m \in \mathcal{L}(X)$, then*

$$\Big\|\sum_{j=1}^m T_j Q_j\Big\| \le C_s\Big(\sum_{j=1}^m \|T_j\|^{s'}\Big)^{1/s'}, \qquad \frac{1}{s} + \frac{1}{s'} = 1.$$

Proof. For $\|x\| \le 1$, the vectors $Q_j x$ have disjoint exterior block supports. Thus

$$\begin{aligned}\Big\|\sum_j T_j Q_j x\Big\| &\le \sum_j \|T_j\|\,\|Q_j x\| \\ &\le \Big(\sum_j \|T_j\|^{s'}\Big)^{1/s'}\Big(\sum_j \|Q_j x\|^s\Big)^{1/s} \\ &\le C_s\Big(\sum_j \|T_j\|^{s'}\Big)^{1/s'}\Big\|\sum_j Q_j x\Big\| \\ &\le C_s\Big(\sum_j \|T_j\|^{s'}\Big)^{1/s'}.\end{aligned}$$

□

Proof of Theorem 1

Assume toward contradiction that $Y \neq 0$ and that

$$\Phi : \mathcal{L}(X) \longrightarrow \mathcal{L}(Y)$$

is a surjective non-injective algebra homomorphism. By automatic continuity [5, 2], Φ is continuous. Its nonzero kernel is a closed two-sided ideal in $\mathcal{L}(X)$, hence contains the finite-rank operators.

For $A \in \mathcal{A}$, put $E_A = \Phi(P_A)$. Each E_A is a nonzero idempotent. Indeed, if $E_A = 0$, then

$$I_X = V_A P_A U_A$$

would imply $\Phi(I_X) = 0$, impossible because $Y \neq 0$. If $A \neq B$, then $P_A P_B$ is finite rank by Lemma 1, so

$$E_A E_B = 0 = E_B E_A.$$

Moreover, for each finite $\mathcal{F} \subset \mathcal{A}$, Lemma 1 gives finite-rank corrections R_A such that

$$\Big\| \sum_{A \in \mathcal{F}} E_A \Big\| = \Big\| \Phi\Big(\sum_{A \in \mathcal{F}} (P_A - R_A) \Big) \Big\| \leq \|\Phi\|.$$

Thus the E_A's are pairwise orthogonal nonzero idempotents with uniformly bounded finite sums.

Let $Y_A = E_A Y$ and let Y_0 be the closed linear span of the Y_A's. We claim that the family $(Y_A)_{A \in \mathcal{A}}$ is boundedly complete. By the open mapping theorem, choose $K_\Phi < \infty$ such that every $T \in \mathcal{L}(Y)$ has a lift $S \in \mathcal{L}(X)$ with $\Phi(S) = T$ and $\|S\| \leq K_\Phi \|T\|$. Applying Lemma 2 to finite disjoint families and then applying Φ, we get a constant C such that for all finite $A_1, \ldots, A_m \subset \mathcal{A}$ and all $T_1, \ldots, T_m \in \mathcal{L}(Y)$,

$$\Big\| \sum_{j=1}^{m} T_j E_{A_j} \Big\| \leq C \Big(\sum_{j=1}^{m} \|T_j\|^{s'} \Big)^{1/s'}.$$

The same rank-one duality argument used in [5, Theorem 1.1] then gives a disjoint lower s-estimate on the decomposition:

$$\|y\| \geq C^{-1} \Big(\sum_{j=1}^{m} \|E_{A_j} y\|^s \Big)^{1/s}, \qquad y \in Y.$$

Consequently, if finite partial sums $\sum_{A \in F} y_A$, $y_A \in Y_A$, are uniformly bounded, then $\sum_A \|y_A\|^s < \infty$, so only countably many terms are nonzero and the net of partial sums converges. This is bounded completeness. Since the finite sums of the coordinate projections E_A are uniformly bounded, bounded completeness implies that Y_0 is complemented in Y.

For every subset $\mathcal{S} \subseteq \mathcal{A}$, bounded completeness gives a projection $Q_{\mathcal{S}} : Y_0 \to Y_0$ onto $\overline{\operatorname{span}}\{Y_A : A \in \mathcal{S}\}$. Distinct subsets give projections at distance at least one, because each Y_A is nonzero. Hence

$$\operatorname{dens} \mathcal{L}(Y_0) \geq 2^{\mathfrak{c}}.$$

Since Y_0 is complemented in Y, also $\operatorname{dens} \mathcal{L}(Y) \geq 2^{\mathfrak{c}}$.

On the other hand, $X = R_\omega^{p,0}$ is separable, so $\operatorname{dens} \mathcal{L}(X) \leq \mathfrak{c}$. A continuous image of $\mathcal{L}(X)$ cannot have density exceeding $\mathfrak{c}$, contradicting the surjectivity of $\Phi : \mathcal{L}(X) \to \mathcal{L}(Y)$. Therefore every surjective homomorphism $\mathcal{L}(X) \to \mathcal{L}(Y)$, $Y \neq 0$, is injective, and X has SHAI.

Finally, $R_\omega^p \cong \mathbb{K} \oplus R_\omega^{p,0}$. The one-dimensional space has SHAI, and SHAI is stable under finite direct sums [3, Proposition 2.9]. Thus R_ω^p has SHAI as well.

Verification and Novelty Notes

The cheap run indexes were searched for '2102.03966', 'SHAI', 'LÎ', 'C_\infty', 'complemented subspace of Lp̂', and BRS-related phrases. The only relevant prior run packet was the separate conditional answer for a $C(K)$ SHAI subcase of [4]; it does not overlap with the present L^p/BRS target. Local later-source search found the concrete first-limit BRS realization and embedding classification in [7] and the general BRS FDD/factorization paper [6]; neither was indexed as an explicit SHAI answer to Johnson–Phillips–Schechtman's BRS subquestion.

The proof above is not a literature-status identification: it combines the Bernoulli-block realization of $R^{p,0}_{\omega}$ with a modified Johnson–Phillips–Schechtman quotient-density argument.

Human review should focus on Lemma 1, especially the assertion that the finite sums of the deleted block-copy projections are uniformly bounded, and on the lower-estimate transfer in Lemma 2.

References

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